

Bivariate SROC Curves

2026-04-17

Introduction

The summary ROC (SROC) curve visualises the relationship between pooled sensitivity and specificity across a meta-analysis of diagnostic studies. Modern SROCs are derived from bivariate or HSROC models that properly account for the correlation between sensitivity and specificity as thresholds vary.

Prerequisites

Sensitivity and specificity; bivariate meta-analysis.

Theory

The bivariate model fits

$$\begin{pmatrix} \text{logit}(\text{Sens}_i) \\ \text{logit}(1 - \text{Spec}_i) \end{pmatrix} \sim N(\mu, \Sigma).$$

The SROC curve is derived by mapping a range of the $\text{logit}(1 - \text{Spec})$ axis back to expected $\text{logit}(\text{Sens})$, given the bivariate mean and covariance. The curve shows how sensitivity trades against specificity as threshold varies.

Assumptions

Logit sensitivity and specificity are bivariate normal; studies vary by threshold; between-study heterogeneity captured by the covariance matrix.

R Implementation

```
library(mada)

set.seed(2026)
df <- data.frame(
  TP = c(45, 60, 32, 80, 28, 55, 70, 50),
  FN = c( 5, 10,  8, 20, 12,  5,  8, 15),
  FP = c(10, 15,  8, 20, 10, 12, 18,  9),
  TN = c(140, 165, 152, 180, 150, 138, 200, 175)
)

fit <- reitsma(df)

# Summary statistics
summary(fit)$coefficients[c("tsens", "tfpr"), ]

# SROC plot
plot(fit, sroclty = 1, sroclwd = 2, predict = TRUE, predlty = 2,
     main = "Bivariate SROC with 95% confidence and prediction regions")
points(fpr(df), sens(df), cex = 0.8, pch = 19, col = "grey40")
```

Output & Results

Pooled sensitivity and FPR estimates; SROC curve with confidence (mean) and prediction (future-study) regions; individual study points overlaid.

Interpretation

“The SROC curve had $AUC = 0.94$, with pooled sensitivity 0.84 and specificity 0.91 at the summary operating point. Individual studies clustered tightly along the curve, consistent with threshold variation rather than diagnostic accuracy variation.”

Practical Tips

- Always display both confidence (mean) and prediction (future-study) regions on the SROC.
- Individual study points should lie close to the curve; large off-curve outliers indicate effect modifiers.
- Do not average sensitivity and specificity independently – their correlation matters.
- For partial-threshold studies (e.g., only low-Sens/high-Spec), restrict the SROC to the covered threshold range.
- Report per PRISMA-DTA with a clear operating-point recommendation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis